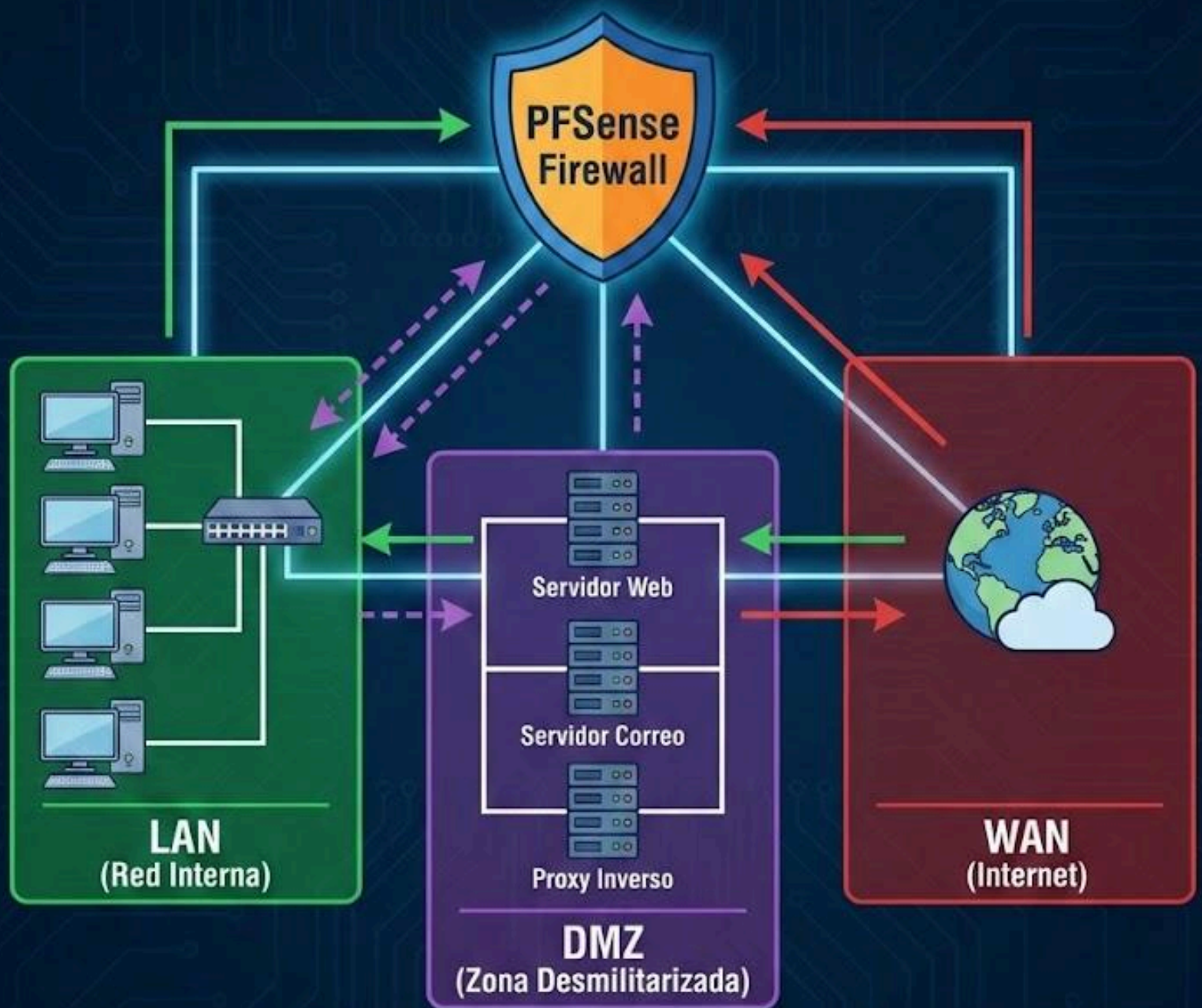


Creación de una DMZ con PFSense

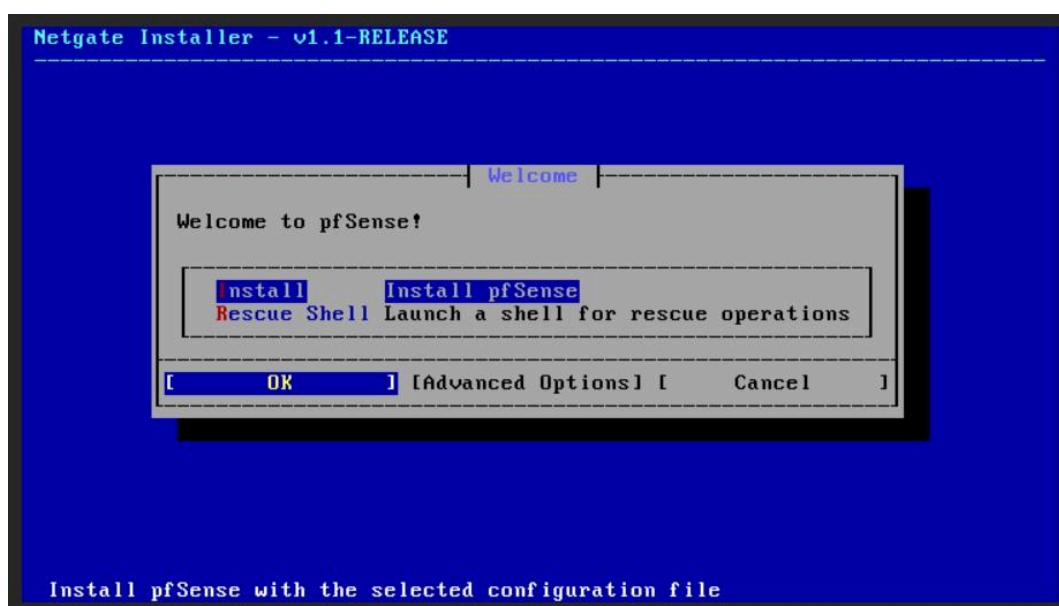


CREACIÓN DE UNA DMZ CON PFSENSE

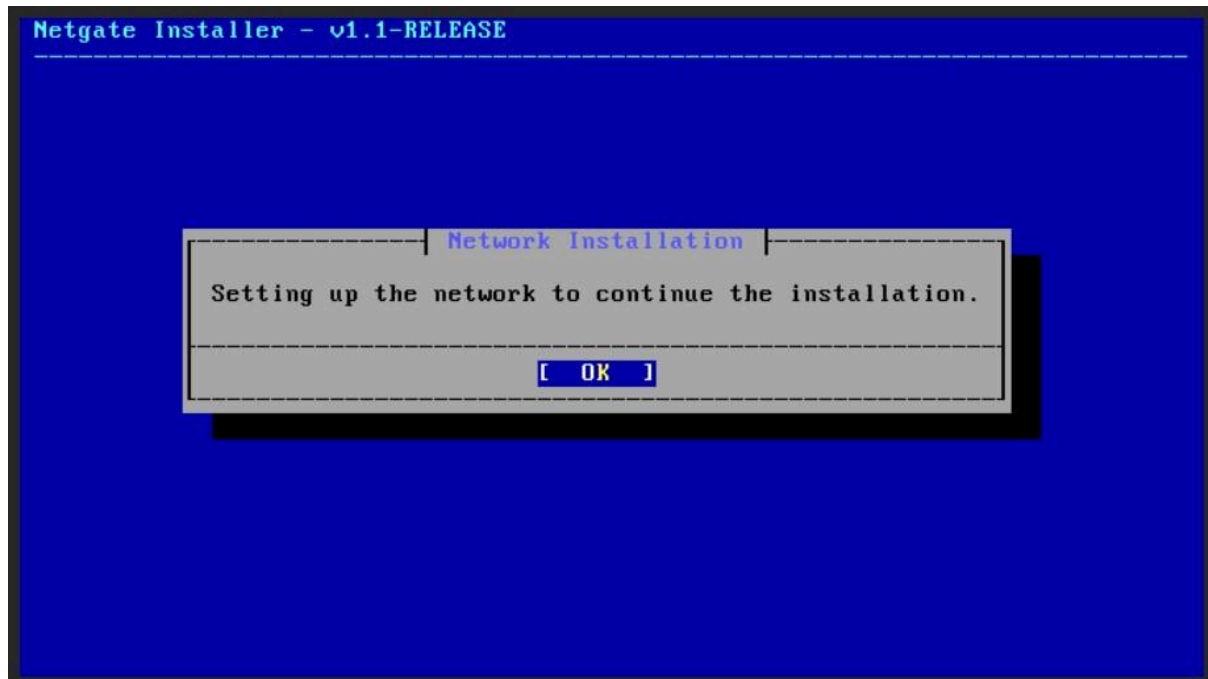
Instalamos PFSENSE en una maquina virtual con 3 interfaces de red

1. WAN (Bridge)
2. LAN (NAT)
3. DMZ (Inactiva al principio)

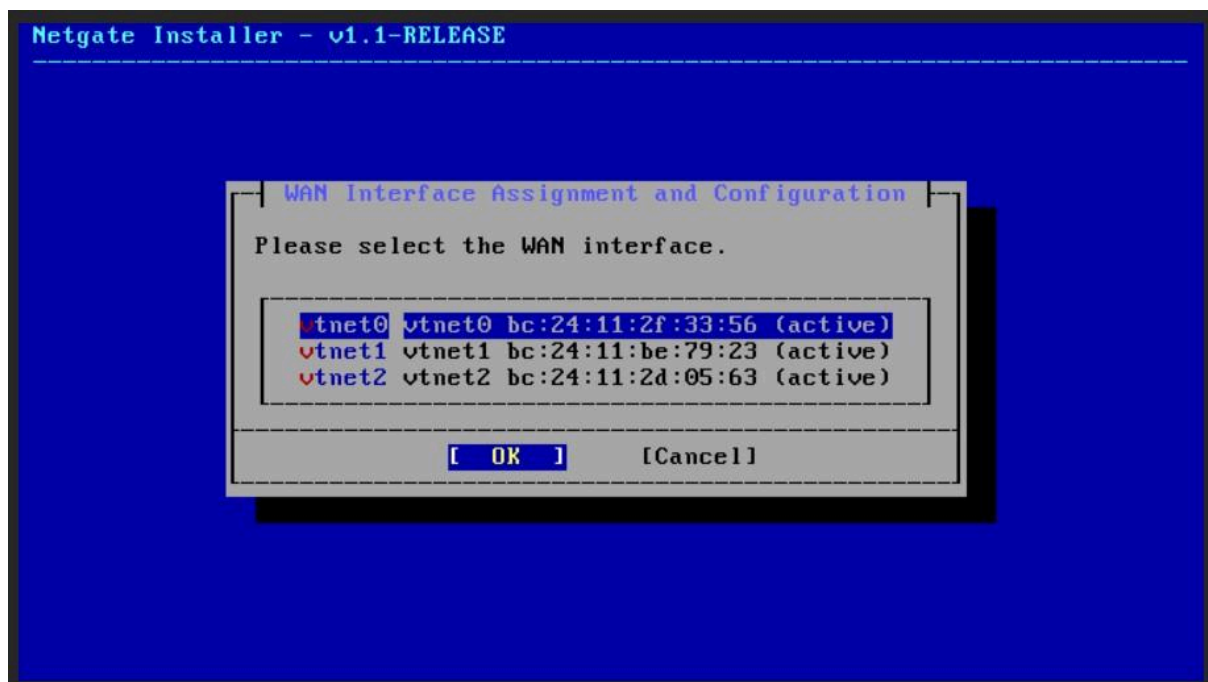
Iniciamos la instalación de Pfsense.



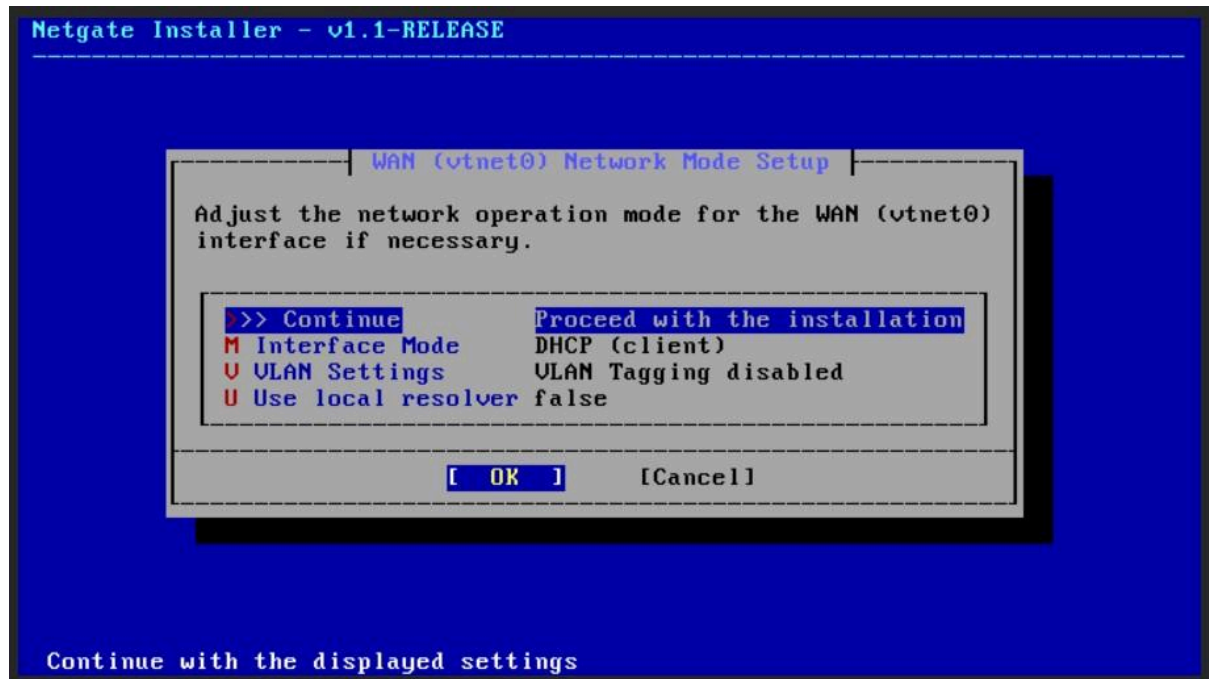
A continuación hacemos la asignación de interfaces para WAN y LAN



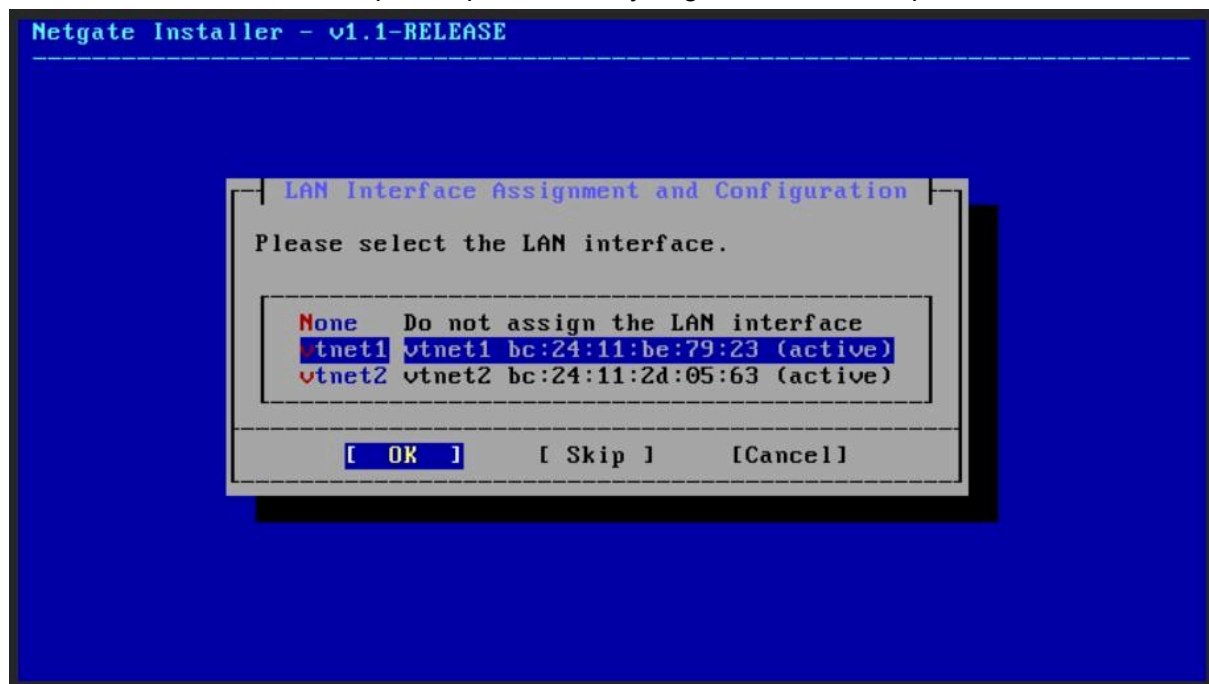
Seleccionamos nuestro adaptador bridge para la WAN, ya que es quien tendra acceso a internet

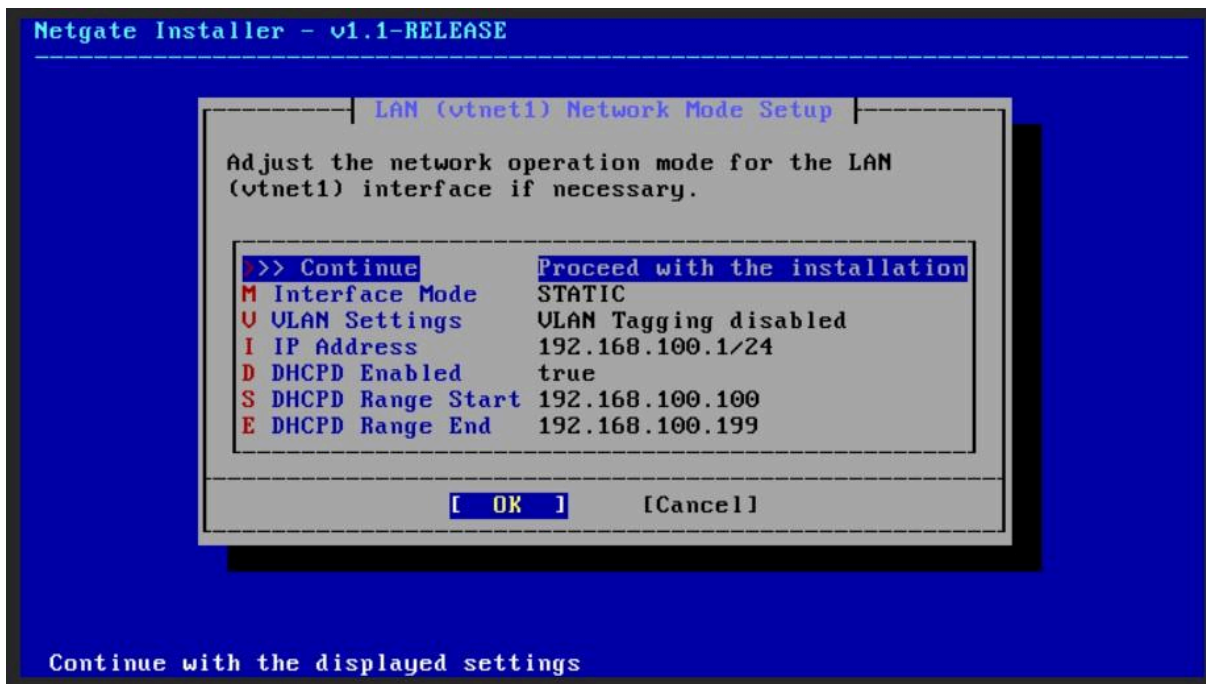


Seleccionamos la configuración deseada para el adaptador y seguimos



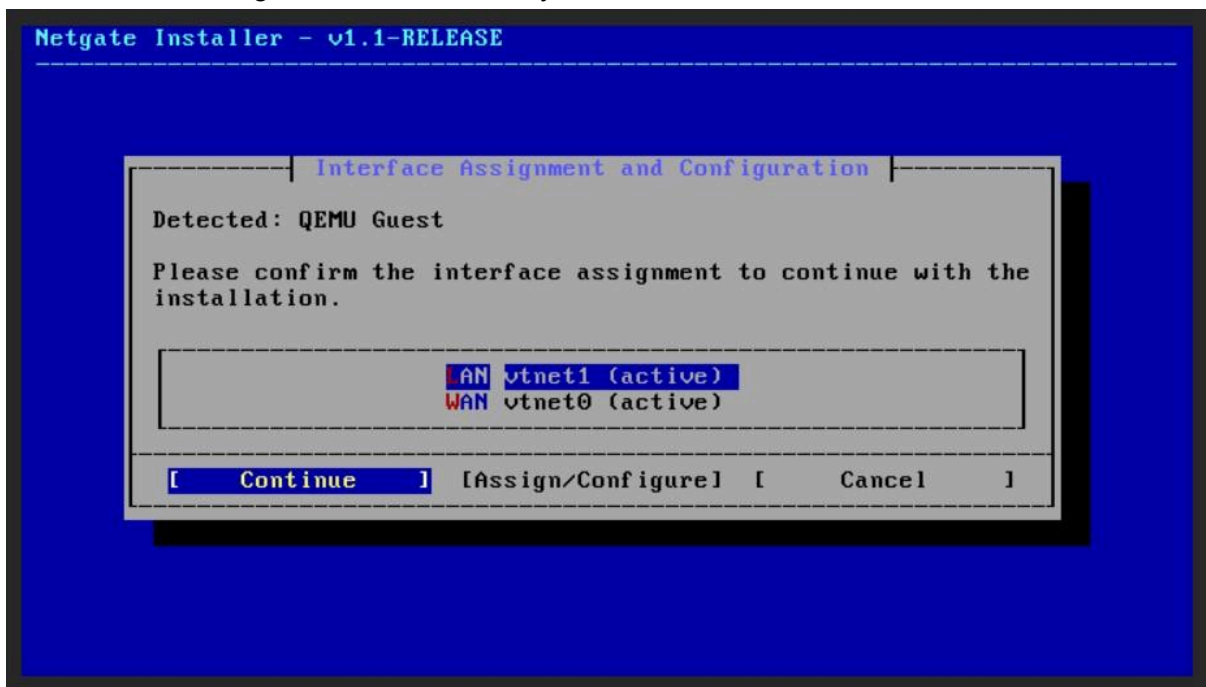
Ahora seleccionamos el adaptador para la LAN y seguimos el mismo proceso





Le damos una dirección estática y creamos una subred para la LAN

Confirmamos la asignación de interfaces y finalizamos la instalación



Netgate Installer - v1.1-RELEASE

Active Subscription Validation

This device does not have an active pfSense Plus subscription.

To purchase pfSense Plus:

Software Store: <https://www.netgate.com/purchase-plus>

NDI: d534 1588 f20e 78de 9a47

and select 'Retry Validation' once purchase is complete.

Or select 'Install CE' to install pfSense CE software.

[Install CE]

[Retry Validation]

[Cancel]

Netgate Installer - v1.1-RELEASE

Installation Options

Please select the File System type and the Partition Scheme.

>>> Continue

Proceed with the installation

F File System

ZFS (recommended default)

P Partition Scheme

GPT (compatible with MBR)

[OK]

[Cancel]

Continue with the displayed settings

Netgate Installer - v1.1-RELEASE

ZFS Virtual Device Type Configuration

Select the ZFS Virtual Device configuration type.

stripe Stripe - No Redundancy

[OK]

[Cancel]

[1+ Disks] Striping provides maximum storage but no redundancy

Netgate Installer - v1.1-RELEASE

Disk Selection

Select the disk(s) for software installation.

[X] a0 32G <QEMU QEMU HARDDISK>

[OK] [Advanced Options] [Cancel]

Netgate Installer - v1.1-RELEASE

Software Version to Install

Select the version of pfSense CE to install.

000-2.8.1 Current Stable Version (2.8.1)
001-2.8.0 Previous Stable Version (2.8.0)
002-2.7.2 Previous Stable Version (2.7.2)

[OK] [Cancel]

Installation Details

/etc/rc.conf. radeonkms for older AMD GPUs can be loaded and there are some positive reports if EFI boot is NOT enabled.

For amdgpu: kld_list="amdgpu"
For Intel: kld_list="i915kms"
For radeonkms: kld_list="radeonkms"

Please ensure that all users requiring graphics are members of the "video" group.

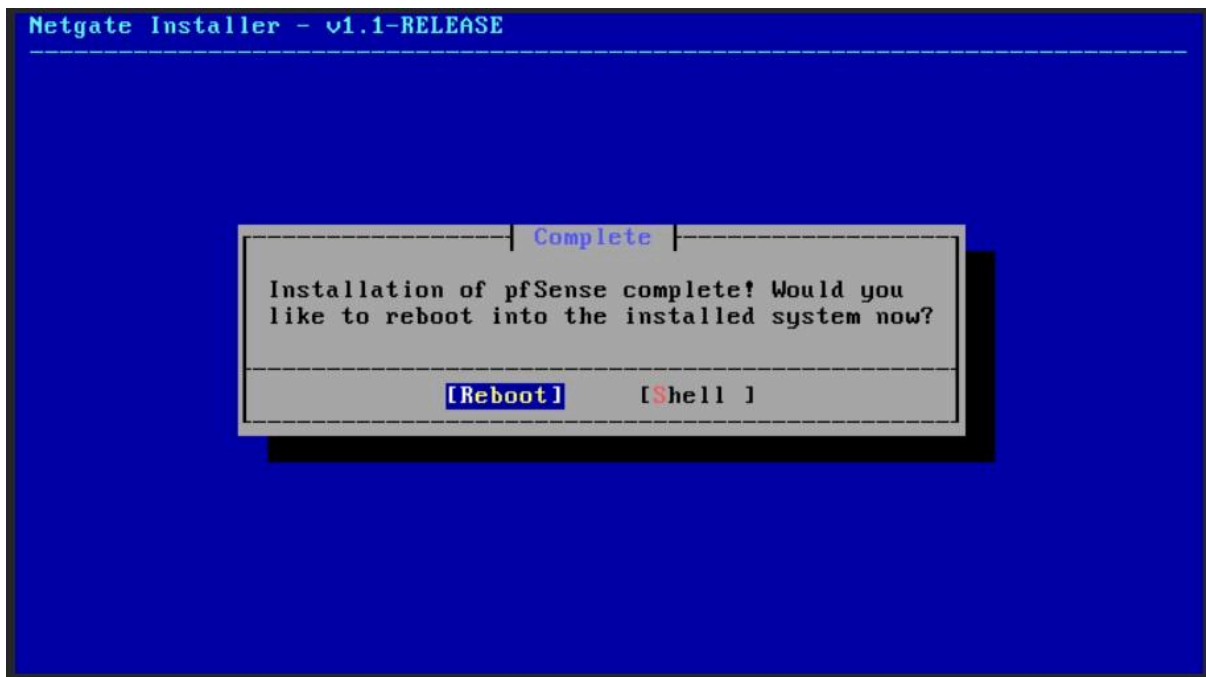
Please note that this package was built for FreeBSD 15.0.
If this is not your current running version, please rebuild it from ports to prevent panics when loading the module.

pfSense Post Installation setup

Exporting the network settings to pfSense.

pfSense Post Installation setup .. done.

< OK >



Una vez completada la instalación solo queda reiniciar la maquina y pfsense estara listo para funcionar

Una vez instalado veremos la siguiente terminal, con la configuración de interfaces y las distintas opciones. Por ahora accederemos a la configuración de pfsense a traves de su interfaz web

```
Starting syslog...done.
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20251126-2112
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

QEMU Guest - Netgate Device ID: d5341588f20e78de9a47

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

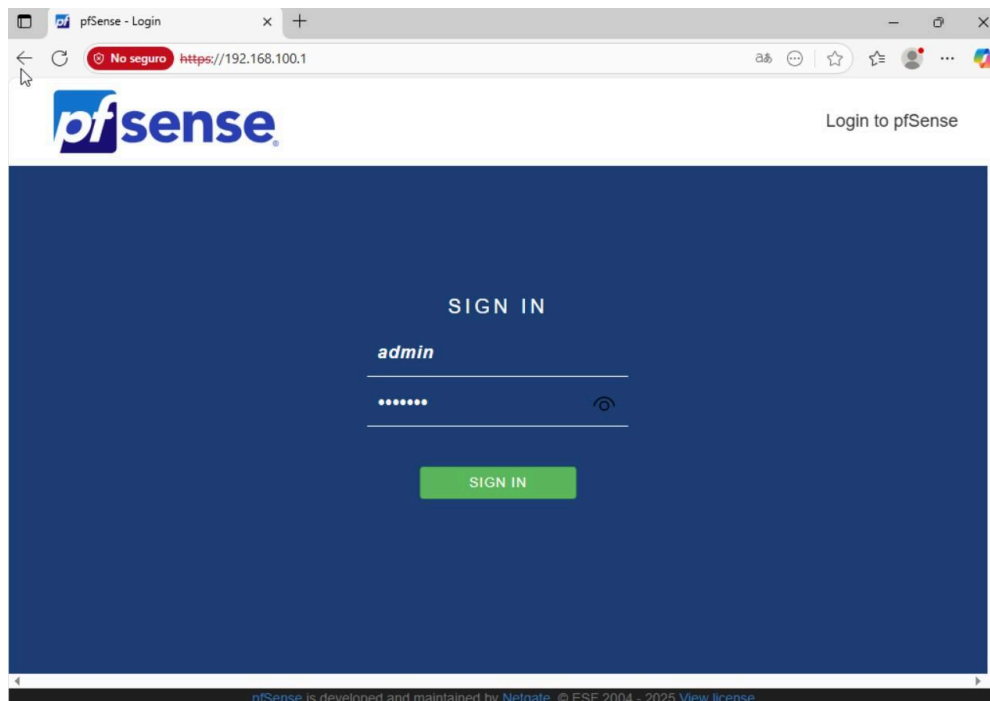
WAN (wan) -> vtnet0 -> v4/DHCP4: 192.168.1.38/24
LAN (lan) -> vtnet1 -> v4: 192.168.100.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: █
```


Configuración de PFSENSE a traves de su interfaz web

Desde otra maquina virtual conectada a la misma LAN que nuestra maquina pfsense, escribimos la ip de pfsense en el navegador para acceder a la interfaz web de configuración



Una vez dentro, accederemos al apartado de interfaces y asignaremos el ultimo adaptador a nuestra DMZ

A screenshot of the pfSense web interface showing the configuration page for the 'OPT1 (vtnet2)' interface. The breadcrumb trail at the top reads 'Interfaces / OPT1 (vtnet2)'. The page is divided into two main sections: 'General Configuration' and 'Static IPv4 Configuration'. In the 'General Configuration' section, the 'Enable' checkbox is checked, and the 'Description' field contains 'DMZ'. The 'IPv4 Configuration Type' is set to 'Static IPv4'. The 'Static IPv4 Configuration' section shows the 'IPv4 Address' as '192.168.200.1' with a subnet mask of '24'. The 'IPv4 Upstream gateway' is set to 'None', and there is a green button to 'Add a new gateway'. A note at the bottom explains that selecting an upstream gateway causes the firewall to treat the interface as a WAN type interface and provides a link to manage gateways.

Interface	Network port
WAN	vtnet0 (bc:24:11:2f:33:56)
LAN	vtnet1 (bc:24:11:be:79:23)
DMZ	vtnet2 (bc:24:11:2d:05:63)

Podemos configurar esta red, como por ejemplo su nombre, que en nuestro caso hemos llamado DMZ

Creación de reglas de firewall para cada interfaz

Reglas en WAN

A la hora de crear reglas seleccionaremos el boton para añadir regla. Una vez hecho esto tendremos acceso al siguiente menu, donde podremos escoger si la reglas es para bloquear o permitir trafico, la interfaz de la regla, el tipo de direccion IP, protocolo, puertos, red, subred, dirección específica y descripción. Este proceso es identico para nuestras 3 interfaces.

Edit Firewall Rule

Action

Block

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (returned to the sender, whereas with block the packet is dropped silently. I

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

WAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

Any

Destination

Destination ☐ Invert match WAN address ▼ Destination Address / ▼

Destination Port Range (other) ▼ (other) ▼

From Custom To Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log ☐ Log packets that are handled by this rule
Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.

Configuraremos las 2 siguientes reglas para nuestra interfaz WAN:

1. Bloquear trafico entrante:
 - a. Acción: Block
 - b. Origen: any
 - c. Destino: WAN
2. Permitir acceso a servicios publicos en DMZ:
 - a. Acción: Pass
 - b. Origen: any
 - c. Destino: Dirección publica WAN/NAT hacia DMZ (Puertos 80/443)

☐	✓	0/0 B	IPv4	*	*	DMZ address	80 (HTTP)	*	none	Permitir HTTP hacia DMZ web server	
☐	✗	0/0 B	IPv4	*	*	WAN address	*	*	none	Bloquear todo el trafico no solicitado	

Asi quedarian nuestra reglas para la interfaz WAN

Reglas en DMZ

3. Bloquear trafico desde DMZ hacia LAN:
 - a. Acción: Block
 - b. Origen: red DMZ
 - c. Destino: Red LAN
4. Permitir acceso desde DMZ a internet (WAN):
 - a. Acción: Pass
 - b. Origen: Red DMZ
 - c. Destino: any
5. Permitir acceso a servicios especificos (comunicación interna DMZ a LAN):
 - a. Acción: Pass
 - b. Origen: Host especifico dentro de la DMZ
 - c. Destino: IP interna del servicio en LAN (Puerto especifico)

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/0 B	IPv4 TCP	192.168.200.1	*	192.168.100.1	80 (HTTP)	*	none	Permitir trafico DMZ host -> Servicio interno en LAN	   
☐	✓	0/0 B	IPv4 TCP	DMZ address	*	*	*	*	none	Permitir salida a internet desde DMZ	   
☐	✗	0/0 B	IPv4 TCP	DMZ address	*	LAN address	*	*	none	Bloquear todo trafico desde DMZ hacia LAN	   

Asi quedarian las reglas para nuestra interfaz DMZ

Reglas en LAN

6. Permitir trafico LAN a internet (WAN):
 - a. Acción: Pass
 - b. Origen: Red LAN
 - c. Destino: any
7. Permitir acceso desde LAN a DMZ (Para administración):
 - a. Acción: Pass
 - b. Origen: Red LAN
 - c. Destino: Red DMZ

☐	✓	0/0 B	IPv4 TCP	LAN address	*	DMZ address	*	*	none	Permitir LAN a DMZ	   
☐	✓	0/0 B	IPv4 TCP	LAN address	*	*	*	*	none	Permitir LAN a internet	   

Asi quedarian las reglas para nuestra interfaz LAN.

Una vez hayamos completado este proceso de configuración podremos guardar los cambios, lo que hara que se empiecen a apicar las reglas y tengamos funcionando nuestra zona desmilitarizada, permitiendo una mayor seguridad dentro de nuestra red.